1.2. **The Equational Theories Project.** As noted in Appendix A, there are 4694 equational laws of order at most 4. The primary mathematical goal of the ETP was to completely determine the *implication graph* for these laws, in which there is a directed edge from E to E′ precisely when $E \models E'$. As the project progressed, an additional goal was added to determine the slightly larger *finite implication graph*, in which there is a directed edge from E to E′ precisely when $E \models_{\text{fin}} E'$.

Such systematic determinations of implication graphs have been seen previously in the literature; for instance, in [59], the relations between 60 identities of Bol–Moufang type were established, and in the blog post [70, §17], some initial steps towards generating this graph for the first hundred or so laws on our list were performed. However, to our knowledge, the ETP is the first project to study such implications at the scale of thousands of laws.

The ETP requires the determination of the truth or falsity of $4694^2 = 22\,033\,636$ implications (for both arbitrary magmas and finite magmas), or $4694 \times (4694 - 1) = 22\,028\,942$ if the reflexive implications $E \models E$ are removed; while one can use properties such as the transitivity of entailment to reduce the work somewhat, this is clearly a task that requires significant automation. It was also a project highly amenable to crowdsourcing, in which different participants could work on developing different techniques, each of which could be used to fill out a different part of the implication graph. In this respect, the project could be compared with a Polymath project [28], which used online forums such as blogs and wikis to openly collaborate on a mathematical research problem. However, the Polymath model required human moderators to review and integrate the contributions of the participants, which clearly would not scale to the ETP which required the verification of over twenty million mathematical statements. Instead, the ETP was centered around a GitHub repository in which the formal mathematical contributions had to be entered in the proof assistant language *Lean*, where they could be automatically verified. In this respect, the ETP was more similar to the recently concluded Busy Beaver Challenge [18], which was a similarly crowdsourced project that computed the fifth Busy Beaver number $BB(5)$ to be 47 176 870 through an analysis of about 180 million Turing machines, with the halting analysis being verified in a variety of computer languages, with the final formal proof written in the proof assistant language *Coq* [68, 18]. One of the aims of the ETP was to explore potential workflows for such collaborative, formally verified mathematical research projects that could serve as a model for future projects of this nature.

Secondary aims of the ETP included the possibility of discovering unusually interesting equational laws, or new experimental observations about such laws, that had not previously been noticed in the literature; and to develop benchmarks to assess the performance of automated theorem provers and other AI tools.

1.3. **Outcomes.** The ETP achieved almost all of its primary objectives, with all of the 22 033 636 implications $E \models E'$ and non-implications $E \not\models E'$ formalized in the proof assistant language *Lean*, and can be found on the ETP GitHub repository. See Figure 1, Figure 2 and Figure 3 for some small fragments of the implication graphs produced. The 4694 laws were organized into 1415 equivalence classes, with by far the largest class being the class of 1496 equations equivalent to the singleton law E2.

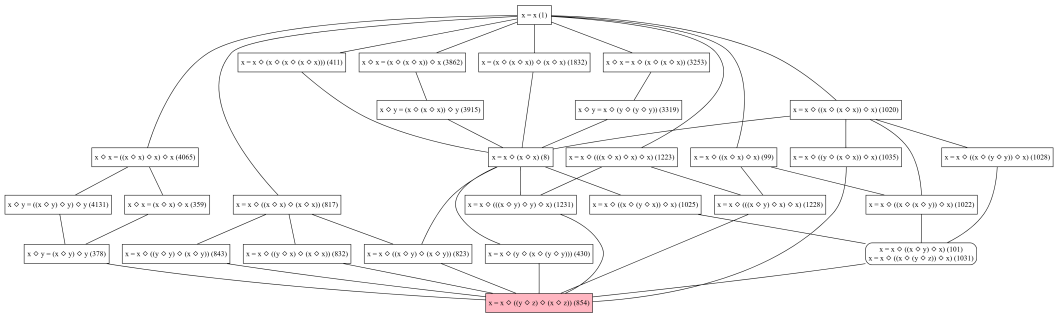

FIGURE 1. A Hasse diagram of all the equational laws implied by E854 (for unrestricted magmas). An edge in this diagram indicates that the lower equation implies the higher one. Rounded rectangles indicate groups of equivalent laws. This graph was produced by the visualization tool *Graphiti*, which was developed for this project.

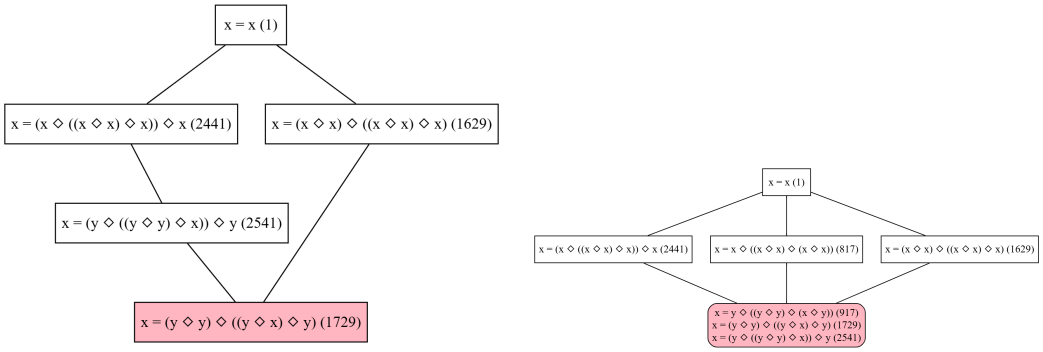

FIGURE 2. A Hasse diagram of all the equational laws implied by E1729, both for unrestricted magmas (left) and finite magmas (right). Note the slightly larger number of implications in the latter.

For the finite implication graph $E \models_{\text{fin}} E'$, we could similarly formalize all but two implications. Specifically, we were unable to obtain either a human-readable or formalized proof or disproof of the implication E677 $\models_{\text{fin}}$ E255 (or its equivalent dual E2910 $\models_{\text{fin}}$ E47), despite extensive efforts from the participants of the project; we tentatively conjecture this implication to be false (i.e., that there exists a finite magma satisfying E677 but not E255), but the refutation appears to be "immune" to most of the techniques that we developed for the project. (We were however able to establish that the corresponding implication E677 $\models$ E255 for arbitrary magmas was false, using the greedy construction discussed in Section 5.5.)

Of the 22 033 636 possible implications $E \models E'$, 8 178 279 (or 37.12%) would end up being true; for an additional set of either 820 or 822 pairs E, E', the weaker implication $E \models_{\text{fin}} E'$ also held. To establish such positive implications $E \models E'$ or $E \models_{\text{fin}} E'$, the main techniques used were as follows:

- A very small number of positive implications were established and **formalized by hand**, mostly through direct rewriting of the laws; but this approach would not scale to the full project.

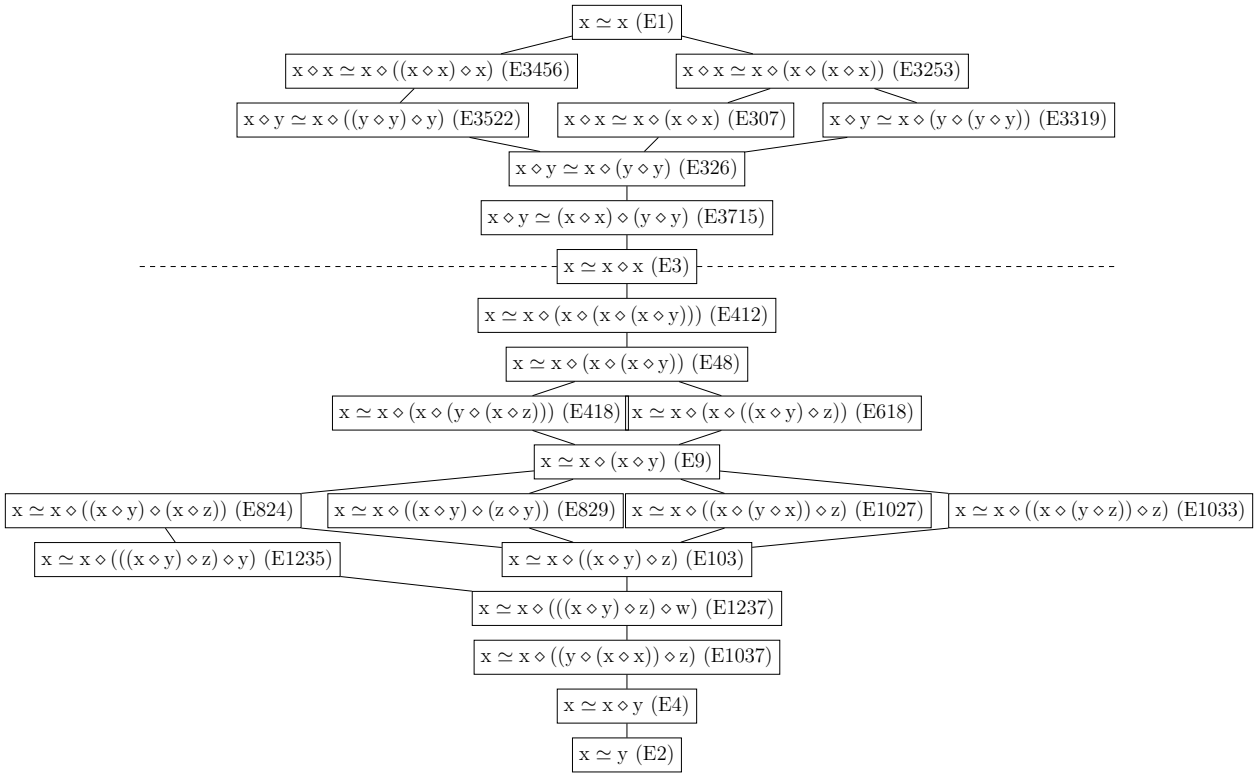

FIGURE 3. Longest chains of implications (length 15) between inequivalent laws in the implication graph. The parts above/below law E3 can be independently dualized.

- **Simple rewriting rules**, for instance based on the observation that any law of the form $\mathrm{x} \simeq f(\mathrm{y}, \mathrm{z}, \dots)$ was necessarily equivalent to the singleton law E2, could already reduce the size of potential equivalence classes by a significant fraction. We discuss this method in Section 6.1.
- The preorder axioms for $\models$, as well as the "duality" symmetry of the preorder with respect to replacing a magma operation $x \diamond y$ with its opposite $x \diamond^{\mathrm{op}} y \coloneqq y \diamond x$, can be used to significantly cut down on the number of implications that need to be proven explicitly; ultimately, only 10 657 (0.13%) of the positive implications needed a direct proof.
- To obtain additional implications for finite magmas, heavy reliance was made on the fact that for functions $f \colon M \to M$ on a finite set M, surjectivity was equivalent to injectivity. Some more sophisticated variants of this idea can lead to additional implications; see Section 5.1.
- **Automated Theorem Provers** (ATP) could be deployed at extremely fast speeds to establish a complete generating set of positive implications; see Section 7.

More challenging were the 13 855 357 (62.88%) implications that were false, $\mathrm{E} \not\models \mathrm{E}'$, and particularly the slightly smaller set of 13 854 535 or 13 854 537 implications that were false even for finite magmas, $\mathrm{E} \not\models_{\mathrm{fin}} \mathrm{E}'$. Here, the range of techniques needed to refute such implications were quite varied, and may be of independent interest: